

ARNAV JOSHI

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Education

Thapar Institute of Engineering and Technology

Aug 2023 – July 2027

B.Tech in Computer Science Engineering

CGPA: 9.57

Ryan International School, Ludhiana

July 2023

Class 12 (Senior Secondary Education)

96.7%

Experience

Samsung R&D Institute, Bengaluru

July 2026 – Dec 2026

Research Intern | Cloud AI/ML

- Building a Retrieval-Augmented Generation (RAG) pipeline for an AI-powered cloud-operations and Kubernetes assistant, enriching natural-language queries with application-specific context.
- Designing an agentic decision layer that lets users accept/reject AI-recommended Kubernetes actions, orchestrating multiple LLM backends into one troubleshooting interface.

Indian Institute of Technology Delhi — FE²B Lab

May 2026 – July 2026

AI/ML Intern | Advisor: Dr. Narsing Kumar Jha

- Designed a feature-engineering and unsupervised clustering pipeline to classify bubble-acoustic flow regimes from 90 real hydrophone recordings, achieving 95.56% cross-validated accuracy.
- Built a physics-informed neural network (PINN-inspired) regression model fusing PIV velocity fields with force-balance priors to predict dipole acoustic pressure and full 360° directivity, validated against 110 real experimental frames.

Projects

SynthSales – Agentic AI Sales CRM | *Python, FastAPI, PostgreSQL, LLM APIs*

GitHub | Live

- Architected an 8-agent LLM pipeline for automated company research, scoring, and outreach, with a confidence-calibrated ranking layer that hard-caps AI-driven scores by up to 54 points when evidence is weak.
- Engineered a 3-provider LLM failover chain (Gemini/Groq/OpenRouter) with automatic rate-limit rerouting, self-correcting structured generation, and a 6-class intent classifier gating replies at a 70% confidence threshold.

Physics-Informed Halo CME Detection System | *Python, XGBoost, Scikit-learn, CDF Ingestion*

GitHub

- Built a physics-informed ML pipeline detecting Halo CMEs from Aditya-L1 SWIS solar-wind data, deriving 4 domain-specific features from 8 candidates via class-separability screening.
- Trained a soft-voting ensemble (Random Forest, XGBoost, Logistic Regression) on 43 labeled event windows, reaching 85% accuracy and zero false negatives; deployed as a public Streamlit app.

Speech Emotion Recognition App | *PyTorch, Wav2Vec2, Streamlit*

GitHub

- Built a deep learning-based speech emotion recognition system using Wav2Vec2 embeddings and handcrafted acoustic features for emotion classification from audio signals.
- Trained on 3 benchmark speech datasets with data augmentation and deployed an interactive inference app reaching 89.1% classification accuracy.

Skills

Programming Languages: Python, C++, JavaScript, SQL, Go

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs, PINNs, Feature Engineering, Signal Processing

NLP & Generative AI: HuggingFace Transformers, Wav2Vec2, LLM APIs, LangChain, RAG, Agentic AI Systems, Stable Diffusion, NLTK

MLOps & Deployment: MLflow, DVC, Docker, Kubernetes, AWS EC2, GitHub Actions, FastAPI, Flask, Streamlit

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, MongoDB, Firebase

Achievements

- Achieved a Top 1500 global ranking in the Google Big Code Competition and was selected as a National Semi-Finalist in Flipkart GRiD 8.0.
- Awarded Merit Scholarship for three consecutive years at Thapar Institute of Engineering and Technology, ranking among the top students of the Computer Science branch.